ECON 340: Economics of the Family
TA Session 8

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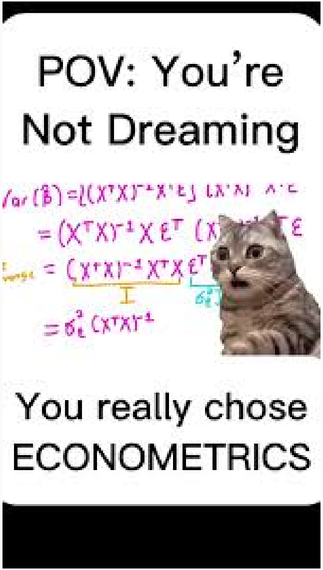
November 2025

Today's Topics



Today

- Difference In Differences and Panel Data
- Instrumental Variables with Panel Data

Motivation

- Ideally, we want to run an experiment to estimate causal effects. Why?
- Often we can't randomly assign people to treatment and control groups.
- We can try to mimic an experiment using observational data.
- But we're worried about unsoverable characteristics – called confounding variables.
- So how should we deal with confounding variables with observational data?

Panel Data

- We observe observations for each unit i (say a person or geography) across multiple periods t
- Why is this useful?

Panel Data

- We observe observations for each unit i (say a person or geography) across multiple periods t
- Why is this useful? It allows us to look at differences in outcomes between treated/untreated units before the treatment occurred
- If treated/control outcomes are different before the treatment, this must be the result of confounding factors.
- So we can potentially use pre-treatment differences to learn about the confounds and adjust for them
- Example: difference-in-differences, which is the most common panel data method used in applied microeconomic research

Example of Diff-in-Diff Setup (Purely Hypothetical)

- Suppose we want to understand the effect of marriage on income (of women)
- Why might marriage affect income?
- We can't randomly assign marriage! So we have to use observational data
- Suppose we only observe post-marriage data on income. What happens if we take the difference of avg. income between married and unmarried folks?
- Omitted variables! Endogeneity! Selection bias!
- In particular, women who choose to marry may have had lower income potential even if they had not married.
- But with panel data, we can actually see what these income differences were before marriage!

- Draw figure

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- Before marriage, say, unmarried women had an average income 20K higher in every period
- Can we assume uncofundedness now?

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- Before marriage, say, unmarried women had an average income 20K higher in every period
- Can we assume uncofundedness now? No!
- A better assumption: the gap would have remained 20K over time, if the married women had not married
- This is the idea behind difference-in-differences (DiD)

- After the marriage, married women have an average income 30K lower
- Under the assumption that the difference should have been 20K, we can attribute the extra 10K difference to marriage
- Thus the treatment effect is the post-treatment difference minus the pre-treatment difference

Formalizing the Assumptions of DiD

- Assume there are 2 periods, $t = 1, 2$. Treated units ($D_i = 1$) are treated in period 2; control units never-treated.
- Let Y_{it} be the observed outcome for unit i in period t. Assume $Y_{it} = D_i\, Y_{it}(1) + (1 - D_i)\, Y_{it}(0)$
- Assumption 1: **No anticipation.** $Y_{i1}(1) = Y_{i1}(0)$.
  - The treatment in period 2 does not affect outcomes in period 1.
- Assumption 2: **Parallel trends.**

$$E[Y_{i2}(0) - Y_{i1}(0) | D_i = 1] = E[Y_{i2}(0) - Y_{i1}(0) | D_i = 0]$$

or equivalently,

$$E[Y_{i2}(0) | D_i = 1] - E[Y_{i1}(0) | D_i = 0] = E[Y_{i1}(0) | D_i = 1] - E[Y_{i1}(0) | D_i = 0]$$

Identification

- Under the assumptions above, we have:

$$
\begin{aligned}
& E[Y_{i2} - Y_{i1}|D_i = 1] - E[Y_{i2} - Y_{i1}|D_i = 0] \\
&= E[Y_{i2}(1) - Y_{i1}(1)|D_i = 1] - E[Y_{i2}(0) - Y_{i1}(0)|D_i = 0] \\
&= E[Y_{i2}(1) - Y_{i0}(0)|D_i = 1] - E[Y_{i2}(0) - Y_{i1}(0)|D_i = 0] \\
&= E[Y_{i2}(1) - Y_{i2}(0)|D_i = 1] + E[Y_{i2}(0) - Y_{i1}(0)|D_i = 1] - E[Y_{i2}(0) - Y_{i1}(0)|D_i = 0] \\
&= E[Y_{i2}(1) - Y_{i2}(0)|D_i = 1]
\end{aligned}
$$

- Thus, the difference in differences identifies the average treatment effect on the treated (ATT):

$$\beta_{ATT} = E[Y_{i2}(1) - Y_{i2}(0)|D_i = 1]$$

- The ATT is the average effect in period 2 for those units that were treated.

Estimating the ATT

- So under the DiD assumptions, we show that the ATT is identified as follows:

$$\beta_{ATT} = E[Y_{i2} - Y_{i1}|D_i = 1] - E[Y_{i2} - Y_{i1}|D_i = 0]$$

- How can we estimate this?

Estimating the ATT

- So under the DiD assumptions, we show that the ATT is identified as follows:

$$\beta_{ATT} = E[Y_{i2} - Y_{i1}|D_i = 1] - E[Y_{i2} - Y_{i1}|D_i = 0]$$

- How can we estimate this? Plug in sample analogues (means, in this case):

$$\hat{\beta}_{ATT} = (\bar{Y}_{12} - \bar{Y}_{11}) - (\bar{Y}_{02} - \bar{Y}_{01})$$

where $\bar{Y}_{dt}$ is the sample mean for group with units $D_i = d$ in period t.

Diff-in-Diff Regression

- Consider the regression:

$$Y_{it} = \beta_0 + \beta_1 Post_t + \beta_2 D_i + \beta_3 (Post_t \times D_i) + \epsilon_{it}$$

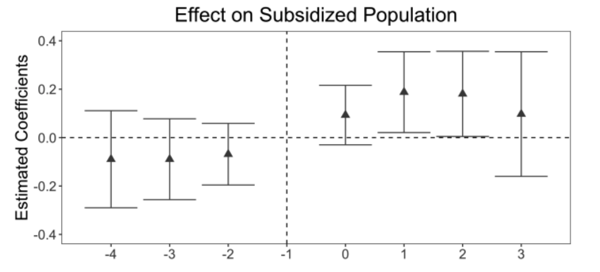
- β_3 is equal to β_{ATT} under the DiD assumptions. How?
- Analogously, $\hat{\beta}_3 = \hat{\tau}_{ATT}$

Parallel Trends

- DiD relies on the parallel trends assumption, which allows for selection bias but requires it to be stable over time. This rules out time-varying confounding factors.
- Often we will be worried about time-varying confounds — e.g., promotions at work for single versus married women in general
- Testing for pre-treatment differences ("pre-trends") can help increase our confidence in the research design. But they're not perfect. Why?
  - Just because trends were parallel beforehand doesn't mean that they would continue to be afterwards
  - Often our estimates of pre-trends are noisy so we're not sure whether they're actually zero or not.

Parallel Trends

- In addition to looking at the point estimates of pre-trends, it's important to consider what the CIs rule out.
- A good rule of thumb for whether a plot is convincing is whether you can draw a smooth line through all the confidence intervals



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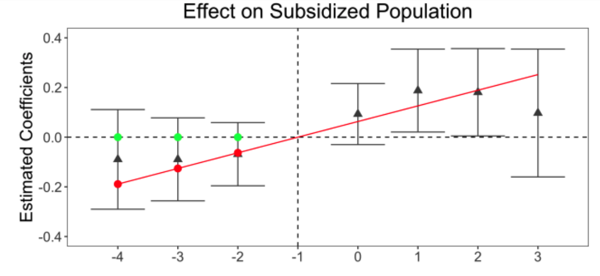
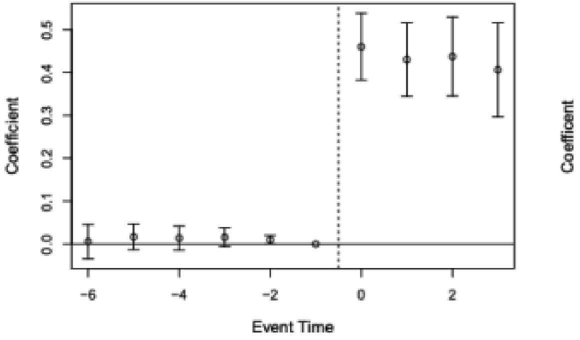


- Is there necessarily a real effect here? Maybe not

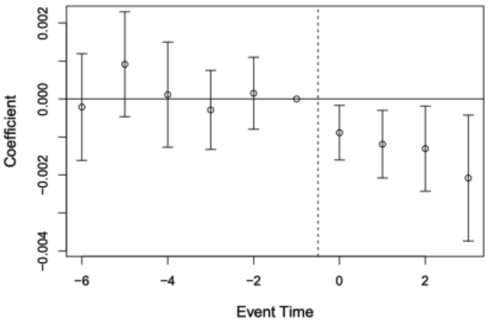
Parallel Trends



(a) Medicaid Eligibility

Parallel Trends

Figure 2: Effect of the ACA Medicaid Expansions on Annual Mortality



Standard Errors in Panel Data

- In standard OLS, we assume that Y_i, X_i are i.i.d. across i. $Y_i = X_i'\beta + e_i$.
- With panel data, we have $Y_{it} = X_{it}'\beta + e_{it}$.
- Is the independence assumption still valid? No! Why not?
  - Now they have to be i.i.d across both i and t.
  - Generally expect Y_{i1} and Y_{i2} to be correlated for the same unit i - serial autocorrelation.
  - If treatment is assigned at a higher level, e.g., state level, then observations within a state will be correlated.

Clustered Standard Errors

- Clustering SEs allows for correlations across observations within a cluster.
- So each cluster is sampled independently.
- Example: with individual clustering, we allow Y_{i1} and Y_{i2} to be correlated for the same individual i.
- But independent for different individuals $i \neq j$.
- Best practice: always cluster at the level of treatment assignment.
- Important: the number of clusters becomes the effective sample size for SEs.
  - So they should be large enough (e.g., > 30).
  - Otherwise, unreliable

Today

- Difference In Differences and Panel Data
- Instrumental Variables with Panel Data

Motivation

- Sometimes we don't have an RCT or even panel data
- Need to worry about other confounding unobservable variables
- In some of these cases, we may need to use instrumental variables (IV) methods
- Basic idea: if the causal relationship between Y_i and X_i is confounded, find a Z_i which "exogenously" shocks X_i
- What does that mean? Back to the Mincer equation!

Mincer Returns to Education

- Let Y_i be earnings, X_i be years of schooling, and ϵ_i be unobserved ability.
- $Y_i = \alpha + \beta X_i + \epsilon_i$
- What's the problem with the OLS estimate of β?

Mincer Returns to Education

- Let Y_i be earnings, X_i be years of schooling, and ϵ_i be unobserved ability.
- $Y_i = \alpha + \beta X_i + \epsilon_i$
- What's the problem with the OLS estimate of β?
- Ability matters! $Cov(X_i, \epsilon_i) \neq 0$

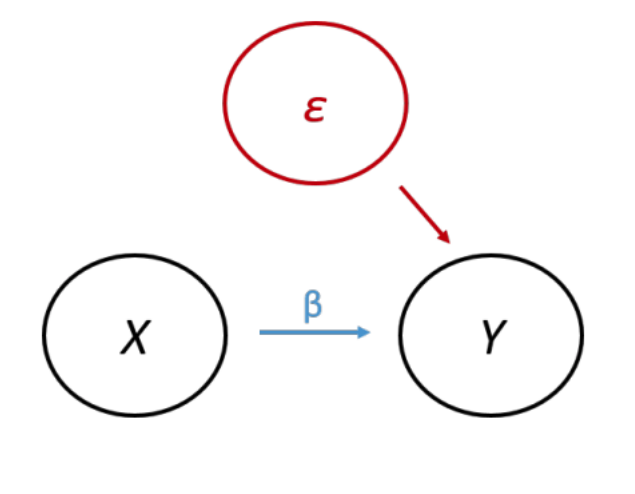
Mincer Returns to Education

- Suppose we observe the outcome of a college lottery that offers scholarship, Z_i.
- If $Z_i = 1$, individual i wins the lottery and gets a scholarship.
- If it truly is a lottery, then in some sense, Z_i is randomly assigned and $Cov(Z_i, \epsilon_i) = 0$. What does that get us?

Mincer Returns to Education

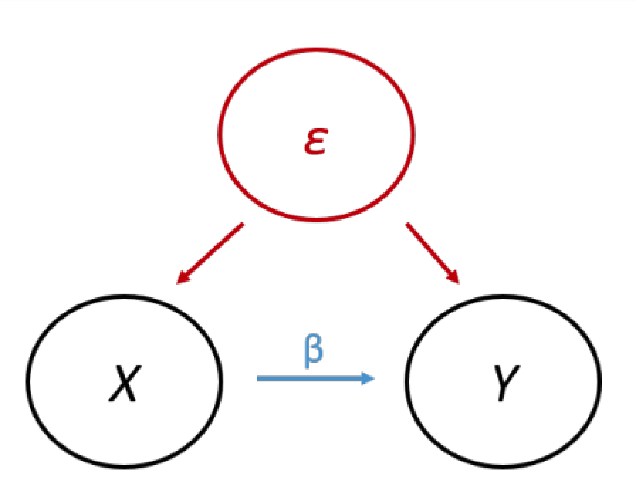
- Suppose we observe the outcome of a college lottery that offers scholarship, Z_i.
- If $Z_i = 1$, individual i wins the lottery and gets a scholarship.
- If it truly is a lottery, then in some sense, Z_i is randomly assigned and $Cov(Z_i, \epsilon_i) = 0$. What does that get us?
- $Cov(Z_i, Y_i) = Cov(Z_i, \alpha + \beta X_i + \epsilon_i) = \beta Cov(Z_i, X_i)$
- So we can estimate β as:
$$\beta = \frac{Cov(Z_i, Y_i)}{Cov(Z_i, X_i)}$$
as long as $Cov(Z_i, X_i) \neq 0$.
- So Z is an instrument for X.

As Good As Random



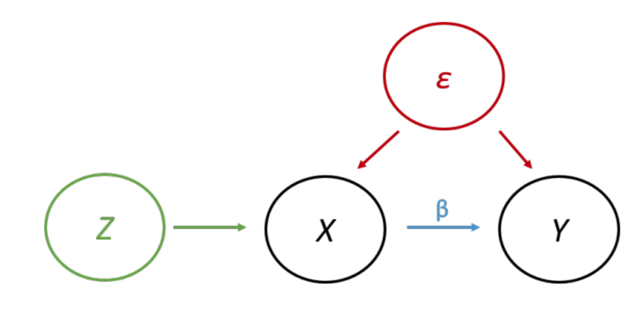
Since $Cov(X_i, \epsilon_i) = 0$, OLS is unbiased and β is causal effect.

Endogeneity



Now $Cov(X_i, \epsilon_i) \neq 0$, OLS is biased and β is not identified.

IV



An instrument with $Cov(Z_i, \epsilon_i) = 0$ and $Cov(Z_i, X_i) \neq 0$ allows us to identify β and addresses endogeneity.

IV Mechanics

- Consider two equations:

$$Y_i = \kappa + \rho Z_i + \nu_i$$
$$X_i = \mu + \pi Z_i + \eta_i$$

- The IV estimand can be written as the ratio of two covariances:

$$\frac{Cov(Z_i, Y_i)}{Cov(Z_i, X_i)} = \frac{Cov(Z_i, Y_i)/Var(Z_i)}{Cov(Z_i, X_i)/Var(Z_i)} = \frac{\rho}{\pi}$$

- We need $Cov(Z_i, \nu_i) = Cov(Z_i, \eta_i) = 0$ for this to be valid.
- With binary Z_i, this is equivalent to:

$$\frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[X_i|Z_i = 1] - E[X_i|Z_i = 0]} = \frac{\rho}{\pi}$$

Heterogeneous Effects

- In previous version, we assumed constant treatment effects.
- β was the same for everyone.
- But effects can be heterogenous across individuals $\implies \beta_i$.
- In particular, when compliance is not perfect, we have:

$$Y_i = \alpha + \beta_i X_i + \epsilon_i$$

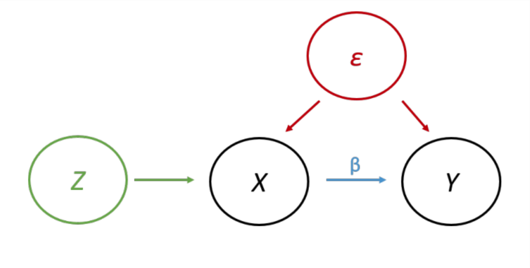
- In this case, IV estimates a Local Average Treatment Effect (LATE).
- LATE is the average effect for compliers – those whose X_i is affected by Z_i.

Validity of an Instrument

- Key validity condition: $Cov(Z_i, \epsilon_i) = 0$.
- Requires two distinct assumptions:
  - As good as random assignment: the instrument must not depend on potential outcomes $Y_i(0), Y_i(1)$ systematically.
  - Exclusion restriction: the instrument affects Y_i only through its effect on the X_i.

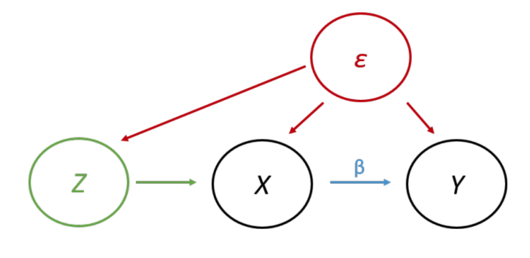
What Makes an Instrument Valid?

Graphically, a DAG can represent the exogeneity condition



What Makes an Instrument Valid?

A violation of as-good-as-random assignment



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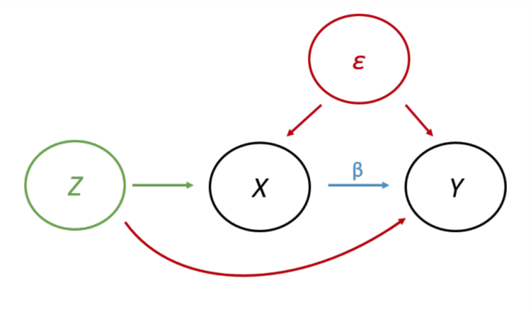
A violation of exclusion restriction



Estimating IV

- With binary Z_i, we can write the IV estimand as a function of population means:

$$\beta = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[X_i|Z_i = 1] - E[X_i|Z_i = 0]}$$

  under the conditions that $Cov(Z_i, \epsilon_i) = 0$ and $Cov(Z_i, X_i) \neq 0$.

- We can estimate this using sample analogues (means):

$$\hat{\beta} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{X}_{Z=1} - \bar{X}_{Z=0}}$$

- More generally, we can use the ratio of the reduced form and first stage coefficients from two OLS regressions:
  - First stage: $X_i = \hat{\mu} + \hat{\pi} Z_i + \hat{\eta}_i$
  - Reduced form: $Y_i = \hat{\kappa} + \hat{\rho} Z_i + \hat{\nu}_i$
  - Then $\hat{\beta}_{IV} = \frac{\hat{\rho}}{\hat{\pi}}$

Quasi-Experimental Evidence

- Sometimes, the instrument can actually be randomly assigned – e.g., lottery
- In other settings, researchers use an instrument that is not explicitly random but may be the result of idiosyncratic factors that are potentially "as-good-as-random"
- This expands the set of cases where we can use IV, but it means the required assumptions deserve extra scrutiny
  - As-good-as-random means Z_i is uncorrelated with background factors relevant for the outcome
  - Remember we still need to gauge exclusion: $Z_i$doesn't affect outcomes except through the specified treatment

Angrist and Krueger (1991)

- AK (1991) are interested in a classic question in labor economics: what are the labor market returns to an additional year of schooling?
- Why can't we just compare the earnings of people with more vs. less schooling?
- What do we need if we want to estimate the effects of schooling on earnings using IV?

Angrist and Krueger (1991)

- AK (1991) are interested in a classic question in labor economics: what are the labor market returns to an additional year of schooling?
- Why can't we just compare the earnings of people with more vs. less schooling?
- What do we need if we want to estimate the effects of schooling on earnings using IV?
  - An IV that is as-good-as-randomly assigned, and affects earnings only through years of schooling

Compulsory Schooling Laws

- Most states have compulsory schooling laws that require people to stay in school until their 16th or 17th birthday
  - At the same time, children tend to start school in September of the year they turn 6
- AK argue this combination of rules will make it so people planning to drop out will have less schooling if they are born earlier in the year
  - That is, if the oldest kid in a class is born on January 1 and the youngest kid is born on December 31, then the oldest kid can legally drop out after getting 1 fewer year of education than the youngest kid
- AK further argue the part of the year in which you are born is effectively random, and therefore instrument for years of schooling with quarter of birth
  - They operationalize this in cohorts born in the 1930s and 1940s, when compulsory schooling laws were more binding

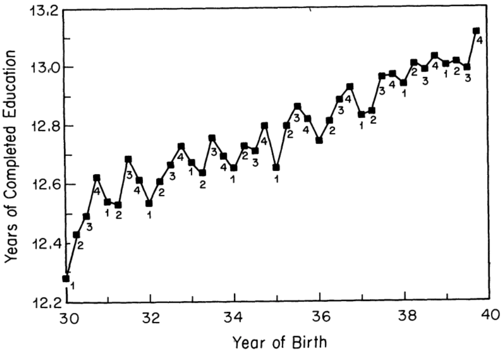
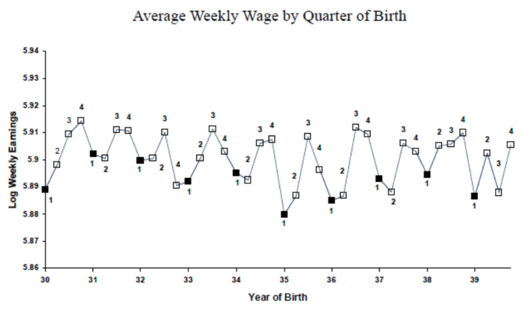


FIGURE I
Years of Education and Season of Birth
1980 Census
Note. Quarter of birth is listed below each observation.

People born in the 1st quarter the of the year tend to have less schooling than people born in 2nd, 3rd, or 4th quarters



Remarkably, people born in the 1st quarter the of the year also tend to have higher earnings than people born in 2nd, 3rd, or 4th quarters

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year
ln (wkly. wage)	5.1484	5.1574
Education	11.3996	11.5252

- What is the first stage estimate?

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- What is the first stage estimate? $11.3996 - 11.5252 = -0.1256$
- What is the reduced form estimate?

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- What is the first stage estimate? $11.3996 - 11.5252 = -0.1256$
- What is the reduced form estimate? $5.1484 - 5.1574 = -0.0090$
- What is the IV estimate of returns to schooling?

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- What is the reduced form estimate? $5.1484 - 5.1574 = -0.0090$
- What is the IV estimate of returns to schooling? $\frac{-0.0090}{-0.1256} = 0.0715$

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) − (2)
ln (wkly. wage)	5.1484	5.1574	−0.00898
			(0.00301)
Education	11.3996	11.5252	−0.1256
			(0.0155)
Wald est. of return to education			0.0715
			(0.0219)
OLS return to education[b]			0.0801
			(0.0004)

- Interpretation: each year of schooling causes a 7.15% increase in earnings
- Why is the OLS different from the IV estimate?

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- IVs tend to have larger standard errors than OLS. Why?

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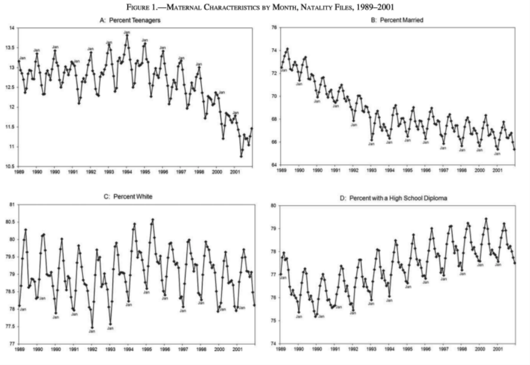
- Interpretation: each year of schooling causes a 7.15% increase in earnings
- Why is the OLS different from the IV estimate? We expect OLS to be biased upwards because of ability bias.
- IVs tend to have larger standard errors than OLS. Why? Because they use less variation in the data.

Evaluating the Assumptions

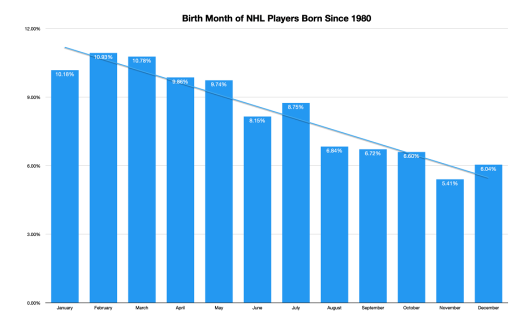
- Relevance: need quarter of birth to be correlated with education. This one can be checked $\implies$ satisfied
- As good as random assignment: Need quarter of birth to be independent of potential outcomes ...
  - Seems plausible IF parents can't time birth dates strategically
  - But...



- Quarter of birth is correlated with some demographic characteristics
- Suggests some possible violations of as-good-as-random-assignment...

Evaluating the Assumptions

- Exclusion: Quarter of birth affects earnings only through number of years of education
- Implies that people whose years of schooling are unaffected by QOB would have same wages if born at different time of the year
- Seems generally plausible but... it turns out being older or younger in your grade may affect the quality of your education directly


Birth Month of NHL Players Born Since 1980
12.00%
9.00%
6.00%
3.00%
0.00%
10.18%
10.93%
10.78%
9.86%
9.74%
8.15%
8.75%
6.84%
6.72%
6.60%
5.41%
6.04%
January
February
March
April
May
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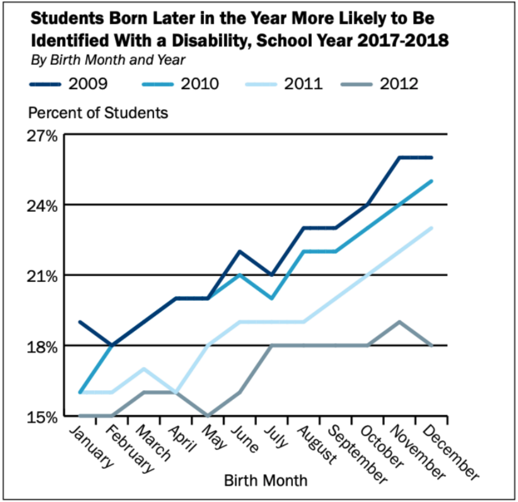
July
August
September
October
November
December


Students Born Later in the Year More Likely to Be Identified With a Disability, School Year 2017-2018
By Birth Month and Year
2009
2010
2011
2012
Percent of Students
27%
24%
21%
18%
15%
January
February
March
April
May
June
July
August
September
October
November
December
Birth Month

Treatment Effect for Whom?

- Suppose that we believe the IV assumptions hold (approximately). How do we interpret the treatment effect we estimate?
- It is a Local Average Treatment Effect (LATE) for compliers — i.e. people who would have gotten an extra year of school if they'd been born in the latter part of the year
- Why might the LATE not correspond with the ATE for the whole population?
- Treatment effects could be different for people on the margin of dropping out ...

Weak Instruments

- Remember, $\hat{\beta}_{IV} = \frac{\hat{\rho}}{\hat{\pi}}$
- If $\hat{\pi} \lim 0$, then $\hat{\beta}_{IV} \lim \infty$!
- This is the *weak instruments* problem - $\hat{\beta_{IV}}$ becomes biased towards the OLS

Testing for Weak IV

- A typical rule of thumb is to worry about many/weak instruments if the F-statistic on the instruments in the first stage is less than 10
- That is, run the first stage regression:

$$X_i = \mu + \pi Z_i + \eta_i$$

- Construct the F-statistic for the null hypothesis that $H_0 : \pi = 0$
- With one instrument, this is just the square of the t-statistic on $\hat{\pi}$
- So $F > 10$ implies $|\text{t-stat}| > \sqrt{10} \approx 3.16$

How to solve the weak IV problem

- Split the sample and use part of the sample to select strong instruments
- Get a larger sample
- Add control variables that correlate with X but not with Z
- Come up with a new instrument! (it's hard tho)